AValidate · Presentation Script

First project presentation, 6 Oct 2026. 15 minutes plus 5 minutes Q&A. Each slide has what to say, the evidence behind it, and questions to expect.

#	Slide	Presenter	Time
1	Cover	Venkat	20 s
2	Problem & Motivation	Swaraj	75 s
3	Project Objectives	Swaraj	60 s
4	Project Scope	Swaraj	60 s
5	Data Required & Available	Ghanashyam	75 s
6	Training Pairs	Ghanashyam	60 s
7	System Design	Venkat	75 s
8	ML Techniques	Venkat	75 s
9	Evaluation Plan	Venkat	90 s
10	What the User Sees	Swaraj	45 s
11	Progress So Far	Ghanashyam	45 s
12	Team & Roles	Venkat	30 s
	Total speaking time		**11 min 50 s**

Per presenter: **Venkat** 4:50 · **Swaraj** 4:00 · **Ghanashyam** 3:00. The rest of the 15 minutes is transitions and buffer. Slides 13 (Thank you) and 14 (Appendix) are not presented.

HOW TO USE THIS

Learn the **Say** part in your own words; do not read it. Learn the **Evidence** so you can defend every claim. Numbers in [brackets] point to the sources at the end. Practise the **If asked** answers aloud, especially the ones for your own slides.

ONE THING TO KNOW BEFOREHAND

AVE has no **siren** or **car horn** class, yet the proposal and slides use dog+siren and car+horn as examples. If asked, say so openly: those are scenario tests run with VGGSound clips. The correspondence score compares embeddings, so it can still flag the mismatch even though the event head cannot name a siren. Prepared answer on slide 4.

01 · Cover

PRESENTER: VENKAT · ABOUT 20 SECONDS

SAY

Good morning. We are Group 25, and our project is AValidate: intelligent audio-visual source matching.
It answers one question: does the thing you see in a video actually make the sound you hear? I am Venkat, with Swaraj and Ghanashyam. Swaraj will start with the problem.

EVIDENCE

- Team and IDs as submitted in the proposal: Veeramani (A0355739M), Swaraj (A0357457N), Ghanashyam (A0357701E). [Proposal V3]

02 · Problem & Motivation

PRESENTER: SWARAJ · ABOUT 75 SECONDS

SAY

The sound in a video often comes from something other than what is on screen.
Look at the four cards. A dog on screen with barking, and a car with a car horn: the visible object is the source, so these are strong matches.
A dog on screen with a siren: the siren comes from somewhere off-screen, so that is a mismatch.
The last one is the interesting case. A person on screen and a dog bark. A dog may be just out of frame, or the person may be imitating a bark. So our system says uncertain, and that is why we report confidence bands instead of a yes or no.
An audio classifier can say 'bark', and a video classifier can say 'person'. Neither tells you whether the two belong together. That gives us our research question: can synchronised audio and visual embeddings be fused to reliably verify whether the visible event is the source of the observed sound?
This is useful anywhere video is searched, moderated or used to train models, for example tagging clips, checking surveillance events, or cleaning noisy audio-visual training data.

EVIDENCE

- The AVE paper defines an audio-visual event as one that is both visible and audible in a segment, and reports that modelling audio and video jointly outperforms modelling them independently. This supports our claim that two separate classifiers are not enough. [1]
- Real-world video is noisy: AVE clips come from YouTube and are described by the authors as unconstrained. [1]
- Learning whether audio and video correspond, using matched and mismatched pairs, is an established self-supervised task ('audio-visual correspondence') introduced in Look, Listen and Learn. [6]

IF ASKED

Q. Isn't this just two classifiers and an if-statement?

That is exactly our baseline. If the audio classifier and the video classifier agree on the label, call it a match. Our evaluation checks whether fusion beats that, especially on similar-sounding mismatches and time-shifted audio, where label agreement fails.

Q. Where would this be used in practice?
Video search and tagging, content moderation, surveillance event checks, and filtering mislabelled clips out of audio-visual training sets. We list these as applications; the project itself is a research prototype.

03 · Project Objectives

PRESENTER: SWARAJ · ABOUT 60 SECONDS

SAY

We have five objectives, in the order the data flows.
One, extract and align: pull synchronised audio and frames from each clip and line them up second by second.
Two, represent: turn each stream into embeddings, first with simple baselines, then with pretrained encoders.
Three, recognise: predict which event it is, and whether it is both audible and visible in each second.
Four, verify: learn a score that says whether what is seen matches what is heard.
Five, present: show the event, the confidence, the per-second evidence and the decision in a simple interface.
The banner is how we will judge success: the fused model must separate matched from mismatched clips better than two independent classifiers whose labels are simply compared, including on hard mismatches.

EVIDENCE

- One-second resolution matches AVE's annotation: start and end times are marked at a resolution of one second. [1]
- These five objectives are the seven MVP bullets of Proposal §1.1, grouped. [Proposal V3]

IF ASKED

Q. Why one-second resolution?
AVE labels every second as event or background, so one second is the finest level we can train and evaluate against. Finer alignment would need labels we do not have.

04 · Project Scope

PRESENTER: SWARAJ · ABOUT 60 SECONDS

SAY

We commit to an offline MVP first, on ten-second clips from AVE.
In scope: the 28 AVE categories, the audio branch, the visual branch, fusion with three heads, a match decision reported as Strong, Weak or Uncertain, or Mismatch, and a demo that takes a clip and shows a timeline.
If time allows: attention-based fusion, scaling up with VGGSound, qualitative localisation on VGG-SS, robustness tests, and a live webcam and microphone mode for the final demo.
And the boundaries. A low score means weak evidence of correspondence; it never proves the pairing is impossible. Events outside the 28 trained categories are not named. Off-screen sources are flagged as uncertain rather than located.

EVIDENCE

- AVE has 28 categories across animals, instruments, home sounds, human speech and vehicles, for example bark, cat, acoustic guitar, banjo, ukulele, mandolin, truck, bus, helicopter, male and female speech. [1][AVE class list]
- Live mode is explicitly an enhancement, not an MVP dependency. [Proposal V3 §5]

IF ASKED

Q. Siren and car horn are on your slides, but they are not AVE categories. How do you test them?
Correct, AVE has neither. The event head will not name a siren, but the correspondence head compares embeddings, so it can still score a dog-plus-siren clip as a mismatch. We will test these scenario cases with clips from VGGSound, which has siren and vehicle classes. Be upfront: this is a scenario test, outside the training categories.

Q. Why not do live mode from the start?
Live mode depends on everything else working. Keeping it as an enhancement protects the MVP if the offline pipeline takes longer.

05 · Data Required & Available

PRESENTER: GHANASHYAM · ABOUT 75 SECONDS

SAY

Our primary dataset is AVE, the Audio-Visual Event dataset. It has 4,143 ten-second YouTube clips across 28 event categories, and every second is labelled as event or background. We use its official split: 3,339 for training, 402 for validation and 402 for testing.
The second set we build ourselves: matched and mismatched pairs, which I will explain on the next slide.
VGGSound is optional. It has over 200,000 clips and over 300 classes, so we can take the classes that overlap with AVE to scale up, or as a fallback. VGG-SS is also optional: about 5,000 clips with bounding boxes around the sound source, which we would only use for qualitative checks.
On the right is the preprocessing pipeline: extract mono audio, sample frames aligned to each second, map the per-second labels, build the pairs, and store versioned splits.
One rule matters most: no leakage. Every clip belongs to one split, and negatives only borrow audio from clips in the same split. Otherwise the model could see test audio during training.

EVIDENCE

- AVE: 4,143 ten-second videos, 28 categories, a subset of AudioSet, split 3,339 / 402 / 402; each video contains at least one audio-visual event of at least two seconds; boundaries annotated at one-second resolution by trained annotators. [1]
- VGGSound: 200,000+ ten-second clips, 310+ classes, including siren classes such as 'ambulance siren'. [2]
- VGG-SS: about 5,000 videos over 220 classes, with bounding boxes on visible sound sources. [3]

IF ASKED

Q. Is 4,143 clips enough for deep learning?
On its own, no, if we trained from scratch. That is why we start from pretrained encoders and only adapt a small number of parameters. Also, 4,143 clips give over 40,000 one-second segments, and generating negatives multiplies the pairs. VGGSound is the fallback if AVE is still too small.

Q. Why AVE and not VGGSound as the primary?
AVE has what we need most: per-second labels saying when the event is both audible and visible. VGGSound has clip-level labels only, so it cannot supervise the per-second head.

06 · Training Pairs

PRESENTER: GHANASHYAM · ABOUT 60 SECONDS

SAY

This is how the model learns what 'matching' means.
A positive is a real clip: a dog and its own barking. Label: match.
An easy negative swaps in audio from a different category, like a siren over the dog. That is easy to reject.
A hard negative uses a similar-sounding category, like a guitar video with banjo audio. AVE has acoustic guitar, banjo, ukulele and mandolin, and also truck and bus, so we have natural hard pairs.
A time-shifted negative keeps the clip's own audio but moves it by a few seconds. Right sound, wrong moment. That tests whether the model uses timing.
Why bother with hard cases? With easy negatives alone, a model can pass just by checking whether two category labels agree. Similar sounds and time shifts force it to use the actual audio-visual evidence.

EVIDENCE

- Swapping audio between clips to create negatives is the standard audio-visual correspondence setup. [6]
- Out-of-sync audio from the same video is an established 'hard negative' that forces models to learn temporal alignment. [7][8]
- Hard-negative pairs available inside AVE: acoustic guitar / banjo / ukulele / mandolin / violin; truck / bus / motorcycle; male / female speech. [AVE class list]

IF ASKED

Q. Two different dog clips swapped: is that a match or a mismatch?
Semantically it is a match, because the visible dog could make that bark, but the timing will not line up. We treat same-category swaps carefully: either exclude them, or count them as time-misaligned negatives only for the temporal head. This is a design decision we will document.

Q. How many negatives per positive?
We will fix a ratio, likely one to one for training, and report it. That is also why we report ROC-AUC and PR-AUC rather than accuracy.

07 · System Design

PRESENTER: VENKAT · ABOUT 75 SECONDS

SAY

Left to right. One clip comes in.
Audio and video are processed in parallel. The audio branch produces one embedding per second, so a ten-second clip gives a ten-by-D matrix. The visual branch does the same, using frames sampled at matching timestamps.
Alignment pairs each second of sound with the same second of video.
The fusion module combines them into one representation per second.
That feeds three heads: which of the 28 events it is, whether it is audible and visible in each second, and the correspondence score from zero to one.
The dashed box is the optional upgrade: cross-modal attention instead of simple fusion.
The banner is our contract as a team. Both branches emit one embedding per second, on the same timestamps, keyed by clip ID. So each branch can be improved without touching the rest.

EVIDENCE

- Separate encoders followed by fusion and a classifier is the architecture used in the AVE benchmark itself, which compares fusion methods including simple concatenation. [1]
- Figure 1 in Proposal V3 was corrected so the two branches are parallel and all three heads come from fusion. [Proposal V3]

IF ASKED

Q. Why late fusion first?
It is the simplest thing that can work, and it gives a clean baseline. If attention does better, we can show by how much. Starting with attention would leave nothing to compare against.

Q. Why three heads instead of one?
They answer different questions and share the same features. The event head and presence head also act as extra supervision from AVE's labels, which can help the correspondence head learn.

08 · ML Techniques

PRESENTER: VENKAT · ABOUT 75 SECONDS

SAY

Every column starts with a baseline, so each improvement is measurable.
Audio: the baseline is a log-Mel spectrogram, which turns sound into an image of frequency over time, fed into a small CNN. The candidate is a pretrained audio encoder such as BEATs, adapted with LoRA.
Visual: the baseline uses frame features from a pretrained image CNN. The candidate is a vision-language encoder such as CLIP ViT-B/16, also adapted with LoRA.
LoRA freezes the pretrained model and trains small add-on matrices, so we can adapt large models on a small dataset and limited GPU.
Fusion starts as concatenation plus an MLP, with cross-modal attention as the upgrade.
The correspondence score is calibrated, then banded into Strong, Weak or Uncertain, or Mismatch, using thresholds chosen on validation data, never on test data.
The named models are indicative; final choices depend on the experiments.

EVIDENCE

- BEATs: self-supervised audio pre-training with an acoustic tokenizer, trained iteratively; reports 50.6% mAP on AudioSet-2M and 98.1% accuracy on ESC-50. [4]
- CLIP: trained on 400 million image-text pairs; zero-shot it matches the original ResNet-50 on ImageNet without using its 1.28 million training images. [5]
- LoRA: freezes the pretrained weights and injects trainable low-rank matrices; on GPT-3 it cut trainable parameters by 10,000 times and GPU memory by 3 times, with comparable quality. [9]
- Technique mapping for the module (from 3 AValidate_Technique_Mapping.docx): supervised learning (event and presence heads); learning from self-generated pairs (correspondence); deep learning (CNN baseline, pretrained transformers, LoRA); hybrid/ensemble (two modalities fused, late vs attention); intelligent sensing (camera and microphone streams aligned and interpreted jointly).

IF ASKED

Q. Why pretrained models and not train from scratch?
AVE is small. Pretrained encoders already know general sounds and images, so we only need to adapt them. LoRA keeps the number of trained parameters small, which reduces overfitting and fits our GPU budget.

Q. Which pattern recognition aspects does the project cover?
All four the module asks for: supervised and self-supervised learning, deep learning, a hybrid of two fused modalities, and intelligent sensing from camera and microphone. At least three are required.

09 · Evaluation Plan

PRESENTER: VENKAT · ABOUT 90 SECONDS

SAY

We fixed five experiments before training anything. The question column tells the story.
One: can each branch recognise the event on its own? Baseline against pretrained encoders, measured with accuracy and macro-F1.
Two, the key experiment: does fusion add anything over two separate classifiers? The baseline runs an audio classifier and a video classifier and calls it a match if the labels agree. Fusion has to beat that.
Three: does it still work when the mismatch is subtle? We split the results by easy, hard and time-shifted negatives.
Four: does it find the right seconds? Per-second F1 against AVE's labels.
Five: can the confidence be trusted? We measure Expected Calibration Error before and after calibration.
We report ROC-AUC and PR-AUC instead of plain accuracy, because accuracy depends on how many negatives we generate. Our headline number is correspondence ROC-AUC on hard negatives, against the label-agreement baseline.

EVIDENCE

- Macro-F1 averages F1 over classes, so rare categories count as much as common ones.
- ROC-AUC: the probability that a random matched pair scores higher than a random mismatched pair; it does not depend on a threshold or on the class ratio. PR-AUC focuses on the positive class and is more informative when classes are imbalanced. [11]
- Expected Calibration Error: bin predictions by confidence and average the gap between confidence and actual accuracy. Modern neural networks are often over-confident, and temperature scaling is a simple fix. [10]
- Why accuracy misleads: with one positive per three negatives, a model that always says 'mismatch' already scores 75%.

IF ASKED

Q. What target numbers do you expect?

We have not set fixed numeric targets yet, because we have no baseline results. The commitment is relative: fusion must beat the label-agreement baseline on hard negatives. Targets will be set once the baselines are in.

Q. What does calibrated mean here?

If the model says 0.8 for a hundred clips, about eighty of them should really be matches. We check that with ECE and a reliability diagram, and fix it with temperature scaling if needed.

10 · What the User Sees

PRESENTER: SWARAJ · ABOUT 45 SECONDS

SAY

This is a mock-up of the demo, not a result.
The user uploads a clip. They see the event, here dog bark, the decision, here strong match with a score, and which confidence band it falls in.
The timeline shows the evidence second by second. Here the dog is visible the whole time but only barks in some seconds, so 'audible and visible' lights up only in those seconds.
The bands come from two thresholds chosen on validation data.

EVIDENCE

- All numbers on this slide (0.91, 0.88, the timeline) are illustrative; no model has been trained yet.
- The demo is planned as a Gradio app; Hugging Face Spaces is a candidate host. [Proposal V1]

IF ASKED

Q. How are the two thresholds chosen?

On the validation set: the upper threshold where precision for 'match' is high, the lower one where precision for 'mismatch' is high. Anything between is reported as uncertain.

11 · Progress So Far

PRESENTER: GHANASHYAM · ABOUT 45 SECONDS

SAY

Design and planning are done. We have the problem and research question, AVE selected as the primary dataset, the system design and heads agreed, and the evaluation plan fixed, including the label-agreement baseline and hard negatives. The proposal was submitted on 30 September, and we have a shared GitHub repository with the work split by modality.
Next, in order: download AVE and preprocess it, build the audio and visual baselines, the pair generator and a shared evaluation script, the first version of fusion, and a first end-to-end run on a single clip.

EVIDENCE

- Repository: github.com/NUS-AIS-GCs/Audio-Video-Confidence (proposal V1 and V3, technique mapping, this deck).
- Update this slide before submission if anything else is finished, for example AVE downloaded.

IF ASKED

Q. What is your timeline?
Data and baselines first, then pairs and the evaluation harness, then fusion, then the demo. Each step produces something we can measure, so we can show progress at every checkpoint.

12 · Team & Roles

PRESENTER: VENKAT · ABOUT 30 SECONDS

SAY

Each of us owns one slice from raw data to model output. Ghanashyam owns audio, Swaraj owns vision and the demo, and I own fusion and verification.
Integration, end-to-end evaluation and one shared inference function are owned by all three of us: a video goes in, and the event, confidence, per-second evidence and match score come out.
Thank you. We are happy to take questions.

EVIDENCE

- Roles match Proposal V3 §4.1 (Member 1 audio, Member 2 vision, Member 3 fusion and verification).

IF ASKED

Q. Who does the data work?
Everyone. Each member prepares the data for their own branch, and the fusion owner builds the pairs. We chose that so nobody is only doing data or only doing models.

Tough Questions, Any Presenter

Q. What is new here, compared with the AVE paper?
The AVE paper localises events in time. We verify correspondence: given what is visible, is it the source of the sound? We add explicit mismatch training with hard and time-shifted negatives, a label-agreement baseline, calibrated confidence bands, and an uncertain outcome for ambiguous cases.

Q. What if the visible object is the right category but silent, and the sound comes from another dog off-screen?
Then the scene is genuinely ambiguous. The per-second head may see the dog and hear a bark, so the score may be high. This is exactly why we never claim certainty, and why the person-plus-bark case is labelled uncertain.

Q. What compute do you need?
Pretrained encoders with LoRA, and features that can be pre-extracted once per clip. That keeps training to a single GPU, for example Colab or the school's GPUs.

Q. How do you stop the model from cheating?
Three ways: negatives only from the same split, hard and time-shifted negatives so label matching is not enough, and thresholds chosen on validation, never on test.

Q. What if pretrained models do not beat the baselines?
That is still a valid result, and the evaluation is designed to show it honestly. We would analyse where they fail by category and negative type.

Q. Why calibrate at all?
Our output is a confidence band. If the scores are over-confident, a 'strong match' would not mean what it says. Calibration makes the bands trustworthy.

Sources

[1] Tian et al., Audio-Visual Event Localization in Unconstrained Videos, ECCV 2018. arxiv.org/abs/1803.08842
[2] Chen et al., VGGSound: A Large-scale Audio-Visual Dataset, ICASSP 2020. robots.ox.ac.uk/~vgg/data/vggsound
[3] Chen et al., Localizing Visual Sounds the Hard Way (VGG-SS), CVPR 2021. robots.ox.ac.uk/~vgg/research/lvs
[4] Chen et al., BEATs: Audio Pre-Training with Acoustic Tokenizers, 2022. arxiv.org/abs/2212.09058
[5] Radford et al., Learning Transferable Visual Models From Natural Language Supervision (CLIP), ICML 2021. arxiv.org/abs/2103.00020
[6] Arandjelović & Zisserman, Look, Listen and Learn, ICCV 2017. (extra reading, not in proposal)
[7] Owens & Efros, Audio-Visual Scene Analysis with Self-Supervised Multisensory Features, ECCV 2018. (extra reading)
[8] Korbar, Tran & Torresani, Cooperative Learning of Audio and Video Models from Self-Supervised Synchronization, NeurIPS 2018. (extra reading)
[9] Hu et al., LoRA: Low-Rank Adaptation of Large Language Models, 2021. arxiv.org/abs/2106.09685
[10] Guo et al., On Calibration of Modern Neural Networks, ICML 2017. (extra reading)
[11] Saito & Rehmsmeier, The Precision-Recall Plot Is More Informative than the ROC Plot on Imbalanced Datasets, PLOS ONE 2015. (extra reading)
[AVE class list] 28 categories as listed in the AVE dataset card, e.g. huggingface.co/datasets/mteb/AVE-Dataset
[Proposal V1 / V3] Documentations/ in the team repository.

Facts [1]–[5] and [9] were checked against the papers or official pages on 5 Oct 2026. Items marked 'extra reading' are well-known background papers outside the proposal's reference list.